3 High energy, high luminosity lepton-hadron colliders [Burak, Frank, Krzysztof, Umit]

The current roadmap for high-energy physics at CERN identifies the Future Circular Collider (FCC) as the most prominent successor to the LHC era. While the FCC-ee is envisioned as the next major step for precision Higgs and electroweak studies, its full-scale operation is projected to commence around the mid-2040s to 2050. This creates a strategic window to fully exploit the LHC infrastructure during the intervening decades. In this context, lepton-hadron colliders emerge as a vital "bridge project", capable of maintaining discovery potential and technological advancement at CERN.

LHeC [2], Stage one [5], LHeC@IP2 [6], LHμC [1], μTRISTAN [7], ultra cold μ beam [8]

3.1 LHeC

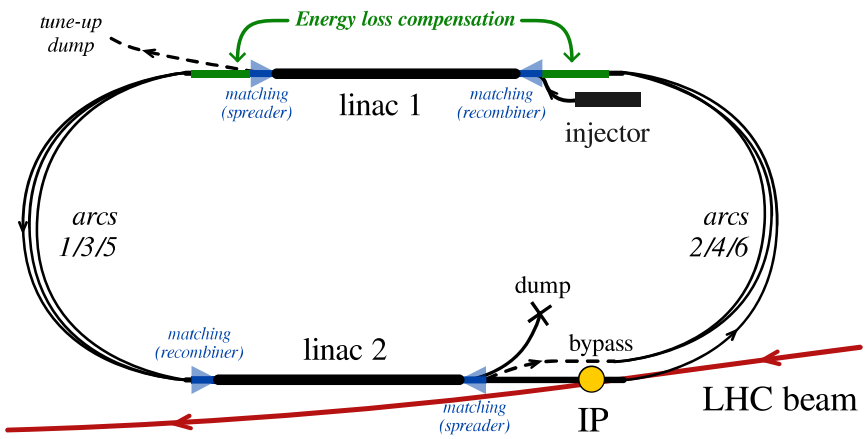

Figure 2. Layout of the proposed LHeC.

3.1.1 Stage-one LHeC

stage-one [5]

3.1.2 Stand-alone LHeC

bridge concept [3]

3.2 LHμC

The proposal for the LHμC arises as an evolution of the LHeC, designed to overcome the energy limitations imposed by synchrotron radiation in the electron sector. By transitioning to muons, it becomes possible to reach multi-TeV center-of-mass energies, which reinforces the argument for lepton-hadron colliders as a vital bridge project at CERN, maintaining its leadership at the energy frontier. However, the deployment of a high-energy muon beam introduces its own set of significant technical challenges. The International Muon Collider Collaboration (IMCC) has been dedicated to extensive R&D about these challenges for a considerable period.

The LHμC framework aims to leverage these global efforts while integrating the recently developed ultra-cold muon production method from J-PARC, instead of the conventional ionization cooling approach. By adopting this ultra-cold muon beam technology, we aim to integrate the subsequent acceleration system within the proposed LHeC layout at CERN. In the following subsections, we will first discuss the processes of high-intensity ultra-cold muon production and explore how the proposed LHeC infrastructure can be utilized for the rapid acceleration of muons. Subsequently, we will outline the design of an optimum main ring and interaction region, tailored to maximize the luminosity of muon-hadron collisions at the LHμC.

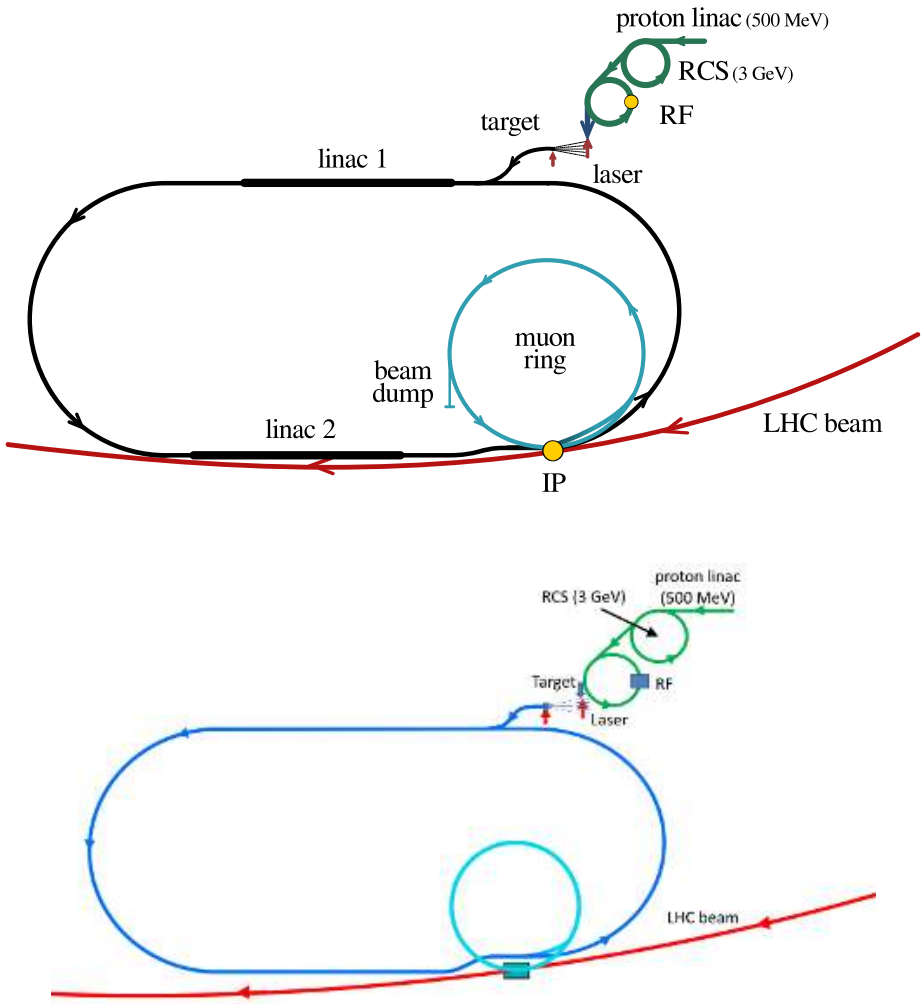

Figure 3. Layout of the proposed LHμC [1], based on the μTRISTAN proposal [7].

3.2.1 Ultra cold μ^+-source

The standard framework for muon production is based on the interaction of a high-power proton driver with a stationary target, typically graphite. This process generates pions and kaons that subsequently decay into muons. Since the resulting muon beam has a very

large phase-space volume, the conventional approach relies heavily on ionization cooling to reduce the beam emittance. As outlined in current Muon Collider design studies [MuCol], this involves a complex and resource-intensive "Front-End" system, including 6D rectilinear cooling and final cooling stages, to prepare the muon beam for acceleration.

However, the design of MC relies on ionization cooling, a complex process that imposes extreme technological demands. Cooling must occur extremely rapidly to counteract the muon's short lifetime, requiring sophisticated wedge absorbers and high-gradient RF cavities operating within intense magnetic fields. Specifically, the final cooling stage requires solenoids capable of generating magnetic fields reaching up to 40 T [HTS Magnets Workshop]. Consequently, the realization of this conventional scheme hinges on the successful development and widespread deployment of advanced High Temperature Superconductor (HTS) magnets. With the construction of a cooling demonstrator scheduled for 2035 at the earliest, these timeline and technological constraints motivate the search for alternatives.

To circumvent these complexities, the LHμC proposal adopts an alternative strategy based on the ultra-cold muon production technology recently advanced at J-PARC. While this method also begins with a proton driver and a primary production target, it diverges significantly by utilizing a laser-ionization process of thermal muonium atom-like structures. The conceptual layout of this muon source and the multi-step production chain of ultra-cold muon beam is illustrated in Figure X.

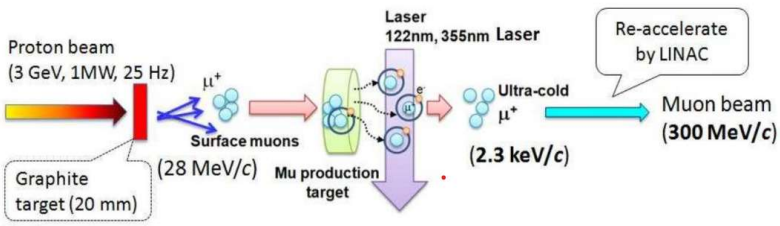

Figure 4. J-PARC ultra-cold muon beam production scheme.

This approach generates an inherently ultra-cold positive muon beam with exceptionally low emittance from the start. The process begins with the generation of "surface muons" from the decay of pions stopped at the near surface of the graphite target. These positive muons are then directed onto a secondary target (typically silica aerogel), where they capture electrons to form thermal muonium (μ^+e^-) atoms. Muoniums diffuse into the vacuum region and are subsequently ionized by high-intensity Lyman-α lasers. The resulting "ultra-cold" muons possess extremely low transverse momentum, yielding a beam with normalized emittance as low as a few mm-mrad. Finally, these ultra-cold muons are injected into a dedicated linac for initial re-acceleration, preserving their low emittance.

In the J-PARC configuration, a single graphite target is used; most produced pions either pass through the target or decay within the material, with only a small fraction of "surface muons" being successfully collected. The low muon rate in this experiment was due to the choice of a laser optimized for spectroscopy rather than ionization; it is not optimized for the high-intensity requirements of a collider. To overcome these requirements,

it is estimated that a muon rate increase of $10^5 - 10^6$ can be achieved at the J-PARC facility through the implementation of a dedicated and optimized production chain [8].

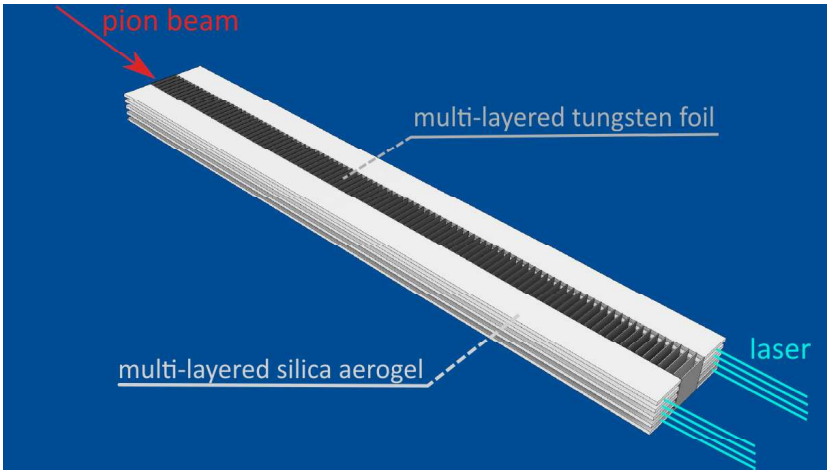

Figure 5. The pion stopping target surrounded by silica aerogel for muonium production and laser ionization, as proposed in the μTRISTAN.

A prominent example of this development is the μTRISTAN proposal, which aims to leverage the J-PARC MLF and TRISTAN infrastructures to provide a roadmap for high-yield ultra-cold positive muon beams in $e^-\mu^+$ and $\mu^+\mu^+$ collider configurations. This enhancement is primarily achieved by implementing a separate pion production target followed by a multi-layered stopping target system, as illustrated in Figure X. Unlike the standard J-PARC setup, the majority of pions that bypass the initial graphite target are transported to a massive 10-meter-long system consisting of approximately one thousand layers of 1 mm-thick tungsten foils. These foils are specifically designed to stop the pion beam while allowing for efficient surface muon production within the inter-layer spaces. These muons then enter a multi-layered silica aerogel structure, where they undergo muonium formation followed by laser ionization to produce the ultra-cold muon beam.

The ultra-cold muons generated across the 10-meter production zone are subsequently collected and focused onto a secondary aerogel target. In this stage, muonium is re-formed and re-ionized by the laser. The two bunches of muons from the pion production ring are combined at the second muon target by tuning the timing of the laser irradiation.

Built upon the J-PARC infrastructure, the μTRISTAN proposal suggests a transformative upgrade to meet the high-intensity requirements of a collider. Unlike the current configuration, where protons strike a target only once, μTRISTAN introduces a pion production ring. In this design, the proton beam is recirculated to strike a thin graphite target up to 40 times before being dumped. This process generates a muon bunch during each turn, resulting in a total of 40 muon bunches per cycle, while an integrated RF cavity compensates for the protons' energy loss in every passage through the target.

When the significant gain in muon collection efficiency, achieved through the multi-layered stopping and muonium production targets, is combined with the 40 bunch and 50 Hz repetition rate, the system is estimated to reach a final muon rate of 10^{13}–10^{14} per

second. This high-flux, ultra-cold source provides the necessary foundation to reach the luminosity goals of the LHμC.

Breakthrough at Muse/J-PARC [8]

3.2.2 μ^+ booster (two scenarios)

vFFA [10] vs. RCS [11]

3.2.3 μ^+ main ring

an optimal circumference is the smallest possible?

3.2.4 Proton ring performance

increasing the proton bunch charge, at lower collisions rate?

3.2.5 Interaction region/MDI

a tentative concept of the beam separation and final focusing

3.2.6 Neutrino flux

3.2.7 1 TeV upgrade

3.2.8 LHμC performance

3.2.9 FCC-μh case

Building on the LHμC experience, a 2 TeV muon beam at the FCC-hh would provide high-luminosity μp collisions at $\sqrt{s} = 20$ TeV!

3.2.10 Outlook/next steps